Identifying and Characterizing Rest Defence Structures using Tracking Data

Abstract

Rest defence (Restfeldsicherung) refers to the players a team leaves behind the ball during attacking possession to guard against counterattacks. Coaches think about rest defence in structural terms: how many players are back, whether they form two lines or one, whether they are marking specific opponents or covering zones, and whether they outnumber the opposition's forwards. These structural properties are well-documented in coaching literature but have not been systematically measured from tracking data. Forcher et al. (2023)[1] quantified rest defence success factors using hand-designed features at the moment of ball loss, and Dash et al. (2025)[2] evaluated the back four's spatial control during defensive transitions, but neither work identifies the full rest defence unit during possession or classifies its organisational type.

This paper presents an unsupervised framework that identifies rest defence structures from tracking data and characterises their configuration. The approach has three stages. In the first stage, we detect when a rest defence structure has formed by looking for a spatial gap in the possessing team's depth profile during opponent-half possession sequences. When a critical mass of players has pushed into the attacking third, the remaining players separate visibly from the attacking group, and this gap serves as both the detection criterion and the split point. Sequences where no clear gap emerges are excluded. In the second stage, we identify the opposition's rest offence players (those positioned as counter-attack threats) using a bipartite coupling matrix that combines time-to-intercept and proximity between rest defenders and opposition players. In the third stage, we identify the role of each defender in the rest defence structure — whether they engaged in man-to-man defending during the sequence or were tasked with protecting the space

We apply the framework to the IDSSE/Sportec Bundesliga open tracking dataset[3]. Centre-backs, defensive midfielders, and the occasional full-back consistently emerge as the rest defence core. Game-level averaged shapes produce distinct profiles that can be compared across teams, and the coupling-based rest offence identification correctly flags opposition strikers and attacking midfielders as the primary counter-threats. In addition, we discuss extensions to temporal dynamics around ball-loss events and connections to transition outcome prediction.

Figure 1 illustrates an example of Fortuna Dusseldorf’s average rest defence structure and rest offence faced across all their games in the dataset. Figure 2 illustrates an example of a

sequence where we identify the rest defence along and characterize the relationship of each defender with the opposition rest offence.

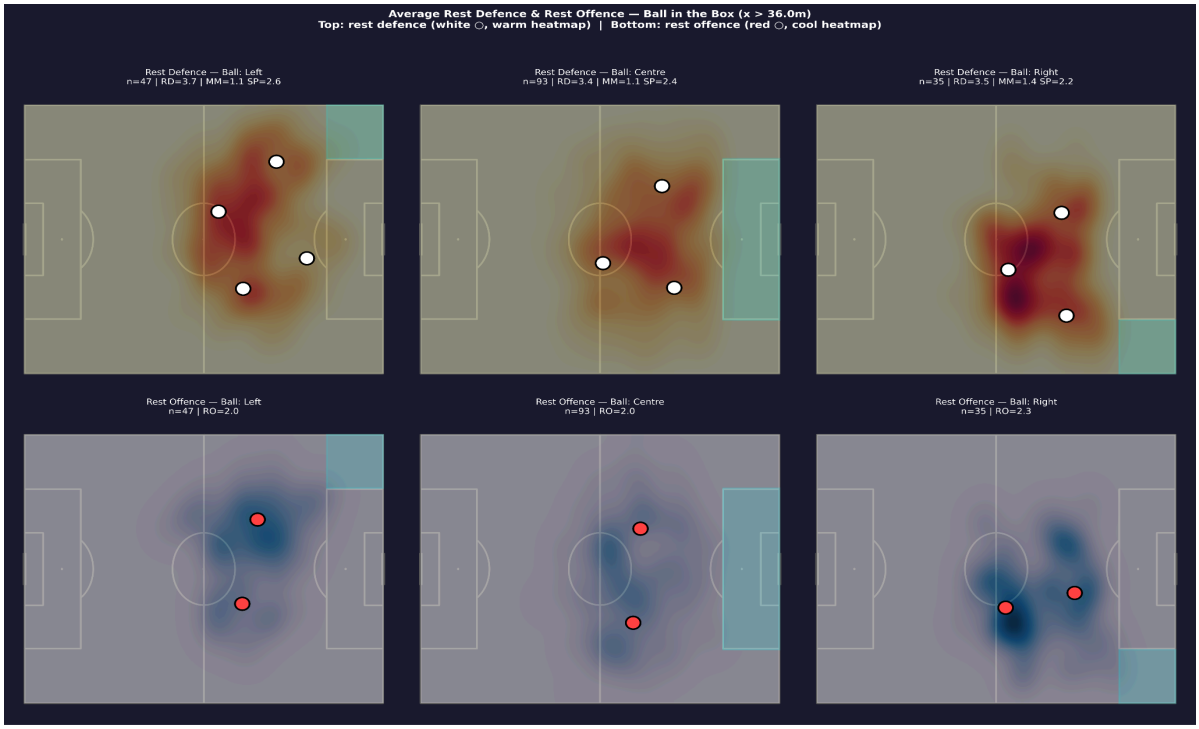


Fig. 1. Fortuna Dusseldorf's average rest defence and rest offence faced based on the ball's position

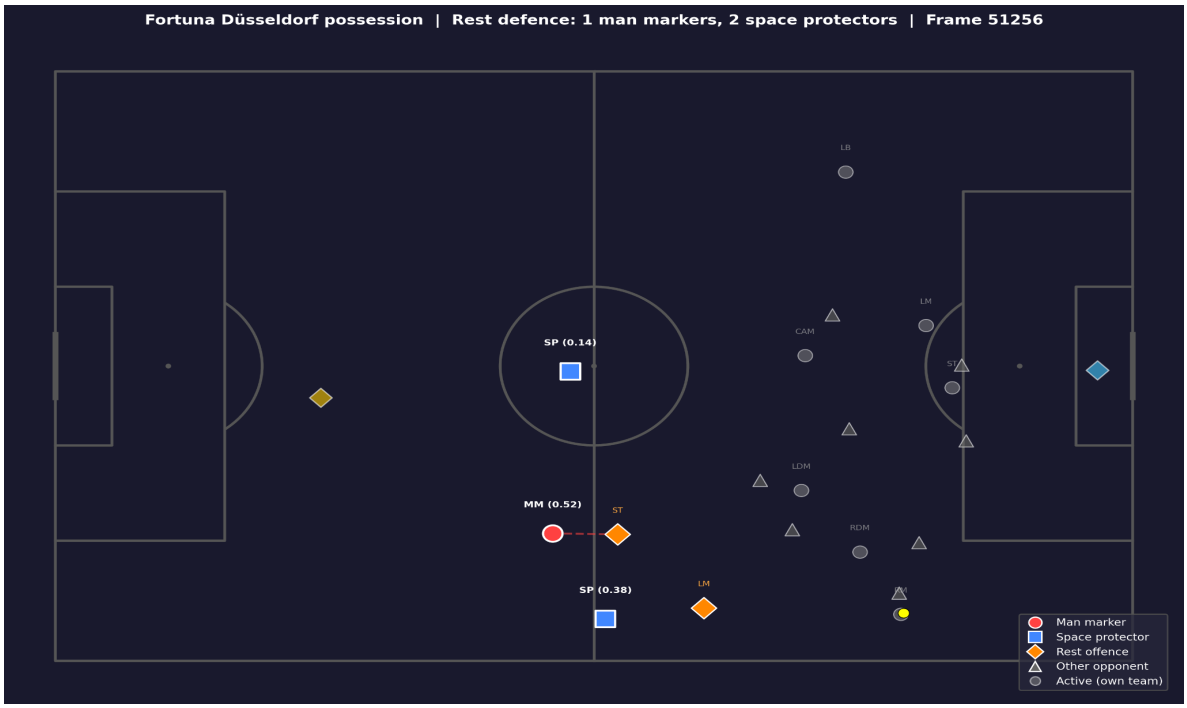


Fig 2. A frame from a sequence illustrating Fortuna Dusseldorf's rest defence structure

References

1. Forcher, L., et al.: A coaching perspective on the success factors of rest defence in soccer. J. Sports Sci. Med. 22, 707–720 (2023)
2. Dash, A., Umemoto, K., Fujii, K.: Prediction-based evaluation of back-four defense with spatial control in soccer. In: ECML/PKDD Workshop MLSA. Springer (2025)
3. IDSSE: International DFL/STS/SciSports/EECS Open Tracking Data Set. Sportec Solutions (2023)